

AI or Not?

Project: Helpful Classroom Sorter

1. AI or Not?

Teach the learning cycle

Primary target: Distinguish an observed result from an explanation of how it was produced

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

You observed spoken time. Speech alone does not tell you whether the message was selected by a timetable, a learned system or a person.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A bus displays a delay prediction learned from past journeys. A written rule turns the display red above 10 minutes. A driver can override the message. Identify all three mechanisms. Would seeing a red display prove that learning occurred?

Concept to explain

Look for evidence of how the system reasons or learns, not just its label or screen.

Term · Meaning

Term: AI system

Meaning: a system using learned patterns or symbolic reasoning

Term: fixed rule

Meaning: an instruction written in advance

Term: human-controlled

Meaning: a person makes the important choice

Term: evidence

Meaning: a clue that supports or changes a claim

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Avoid making AI a yes-or-no property of a whole product before defining the component and decision. Modern learned models and symbolic reasoning both belong in the broader AI discussion. The learning goal here is evidence-based mechanism classification, not memorising a list of smart objects. Do not infer training from responsiveness, language, movement or one correct output.

Readiness answer

You observed spoken time. Speech alone does not tell you whether the message was selected by a timetable, a learned system or a person.

The bus diagram and photo-library case from the student chapter

Blank paper and three colours of pencil

A copy of workbook tasks W1 to W5; no online accounts or real student data

End goal for the pictorial case

By the end, I can separate a learned prediction, a written rule and a human decision.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT a command, signal, photo, sound, or number

PROCESS a human choice, fixed instruction, or learned pattern

OUTPUT movement, calculation, action, or prediction

CORE IDEA Look for evidence of how the system reasons or learns, not just its label or screen.

UNDER THE HOOD A system should be classified by its decision mechanism. Human control, fixed automation, and model-based prediction can produce similar visible behavior, so appearance and marketing words are weak evidence.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE A remote-control car, calculator, motion light, smart speaker, photo sorter, and robot vacuum may

all look clever, but they do not make decisions in the same way.

STEP 1 — OBSERVE a command, signal, photo, sound, or number

STEP 2 — TRACE a human choice, fixed instruction, or learned pattern

STEP 3 — CONCLUDE movement, calculation, action, or prediction

MODEL REASONING Follow the information and identify who or what makes the important choice. A robot body, screen, voice, or the word smart cannot prove AI. If two mechanisms fit the evidence, say that more information is needed.

DO NOT OVERCLAIM Do not call a system AI because it looks, moves, or talks like a person.

What new evidence could change this answer?

Teacher precision note

This course concentrates on modern machine learning. AI also includes symbolic and rule-based approaches. A simple automatic switch or robot body is not, by itself, evidence of AI. Discuss how the system reasons or learns, rather than judging appearance.

Concrete teaching case

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Classify all six using mechanism evidence. Which two need more information?

Exact explanation to model

The lamp uses a fixed rule; the photo sorter uses a learned model; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need mechanism details. Products can combine mechanisms. A robot body or smart label proves none of them.

Explain the added worked example

The school bus display says DELAY 12 MINUTES in red. Leila calls it AI because it predicted the delay. Arun calls it a simple rule because the number turned red. Both have noticed a real part of the product, but neither has explained the whole system. To investigate, ask which decision you are talking about and what evidence reveals how that decision was made.

An output is what a system produces. A mechanism is how it produces that output. Seeing a red number establishes its colour; it does not reveal where the number came from. Similar outputs can come from different mechanisms, and one product can contain several mechanisms.

For this lesson we have an engineering note: the delay estimator was fitted using records of earlier bus journeys; a person wrote the colour instruction; and the driver can replace the message. These are supplied facts for our classroom case. In a real investigation, we would need trustworthy documentation, a demonstration of the process or another source of mechanism evidence.

In the first row, fitting means adjusting a model using examples. Applying the fitted model to a new journey is a different step. The model can produce an incorrect estimate even if it was trained carefully. In the second row, a written threshold compares the estimate with 10. It can follow its instructions perfectly while using a bad estimate. In the third row, a person makes the message decision.

AI is a broader field than learning from examples. It also includes approaches that represent knowledge and reason with rules. Therefore, do not use the shortcut that every rule is non-AI. In this case, a simple colour switch is ordinary automation; identifying that switch does not classify every other part of the bus system.

Key terms

Term · Meaning

Term: Mechanism

Meaning: The process that produces a result

Term: Written rule

Meaning: An instruction specified by a person before the system applies it

Term: Learned model

Meaning: A system whose adjustable values were fitted using examples

Term: Evidence

Meaning: Information that supports or challenges a particular claim

Term: Hybrid system

Meaning: A product that combines more than one kind of process

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Question: Does the red display prove that a learned model was used? First separate the claim from the observation. The observation is that the displayed number is red. The claim concerns the process that produced the number. Next find relevant evidence. The engineering note says earlier journeys were used

to fit the delay estimator. That note supports a learned-model classification for the estimator. The red colour alone does not.

The estimator uses a learned model · The note describes fitting on past journeys · Supported for the estimator

The estimator uses a learned model: The red colour proves learning

The note describes fitting on past journeys: Only the displayed colour is observed

Supported for the estimator: Not supported by colour alone

The estimator uses a learned model: The driver override is a model prediction

The note describes fitting on past journeys: A driver selects the replacement message

Supported for the estimator: Contradicted in this case

A strong answer states its limit: I can identify these three mechanisms from the supplied note, but I cannot infer the model's accuracy from one estimate. Even if the bus arrives exactly 12 minutes late, one success does not establish that the estimator will always work.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

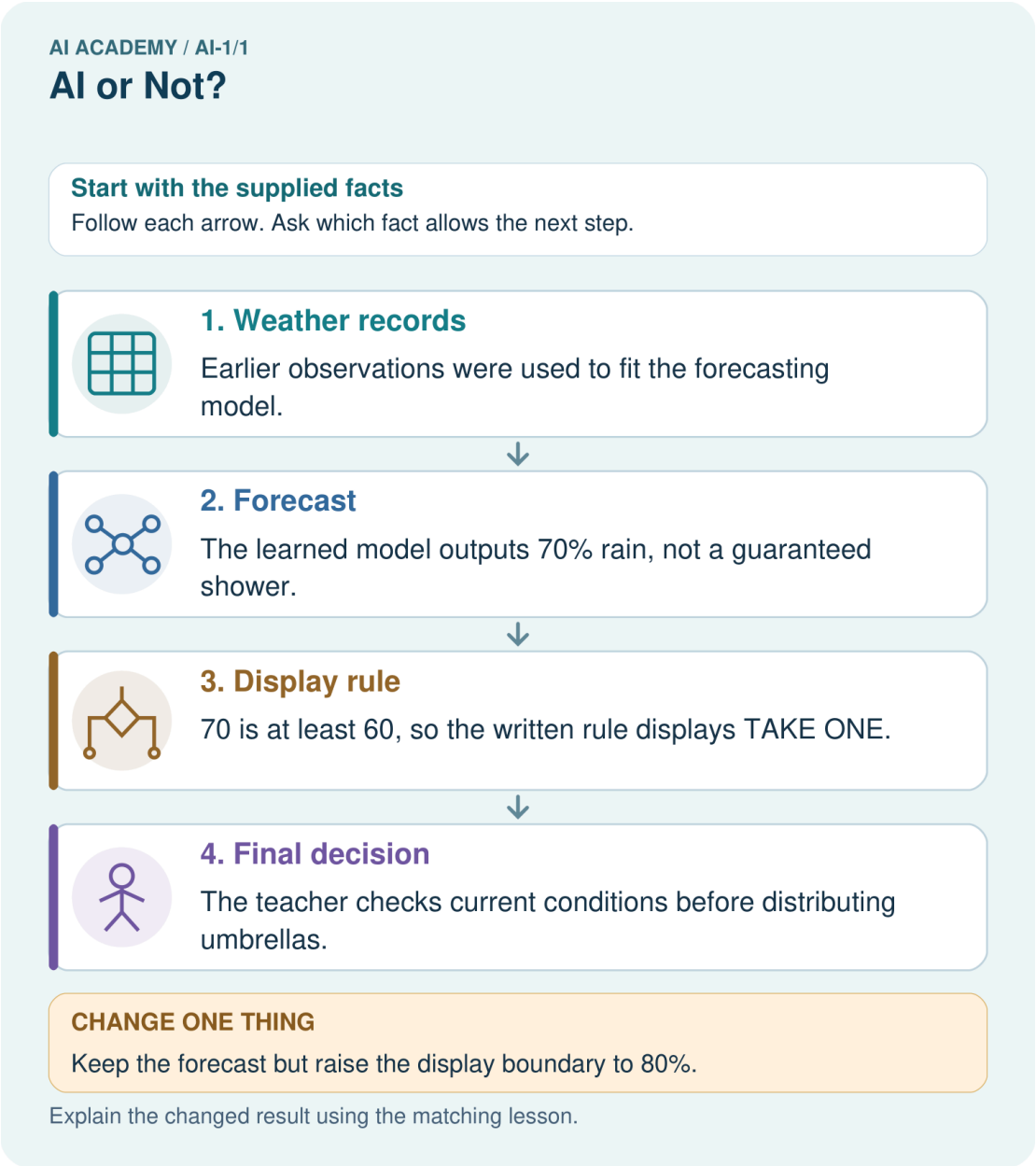
Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: AI or Not?

A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

The facts we will use

A weather model was fitted on earlier weather records; its forecast is 70% rain.
A written rule shows TAKE ONE at rain forecasts of 60% or more. The teacher decides whether to distribute umbrellas.



Read the picture one step at a time

Step · What happens and why

Step: 1. Weather records

What happens and why: Earlier observations were used to fit the forecasting model.

Step: 2. Forecast

What happens and why: The learned model outputs 70% rain, not a guaranteed shower.

Step: 3. Display rule

What happens and why: 70 is at least 60, so the written rule displays TAKE ONE.

Step: 4. Final decision

What happens and why: The teacher checks current conditions before distributing umbrellas.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The fitting record supports calling the forecasting stage learned. Comparing 70 with 60 is an explicit rule. The final action belongs to a person. Merely seeing TAKE ONE would not reveal any of these mechanisms; the process notes supply that evidence.

What this example does not establish

A correct rule comparison does not establish that rain will occur or that 70% forecasts are followed by rain on about 70 out of 100 comparable days.

Change one thing in the pictured case

Change: Keep the forecast but raise the display boundary to 80%.

Prediction: The display no longer says TAKE ONE.

Reason: 70 is below 80; changing a display rule does not refit the model.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Write the question to explain: Classify all six using mechanism evidence. Which two need more information?

Reveal after predictions: The lamp uses a fixed rule; the photo sorter uses a learned model; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need mechanism details. Products can combine mechanisms. A robot body or smart label proves none of them.

Keep for an independent attempt: A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a learned model? What evidence supports each classification?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Prediction stage is learned because labelled examples were used to fit the model; album placement follows the stated written rule; the owner move is human control. Inspect the predicted label and the album action record. A correct answer from a speaking device does not identify the mechanism because recordings, rules, learned models and human messages can all produce speech.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 8 minutes

Teacher action: Ask the clock prerequisite question. Record observed and inferred statements in separate columns. Read two examples aloud without praising a guess as a fact.

Stage: 8 to 23 minutes

Teacher action: Read the bus case. Trace each row of the diagram with a finger. Ask which decision each row makes. Say that a product can combine mechanisms.

Stage: 23 to 35 minutes

Teacher action: Model the red-display question using claim, relevant evidence and conclusion. Cover the final table row and ask students to justify it.

Stage: 35 to 45 minutes

Teacher action: Pairs complete the photo-library explanation and W1. Listen for a distinction between a predicted label and an album action.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Retrieve the distinction with the speaking-device exit question. Revisit the diagram if students rely on appearance.

Stage: 8 to 23 minutes

Teacher action: Complete W2 and W3 in pairs. Each partner must offer a different possible cause of a misplaced photo or direction change.

Stage: 23 to 38 minutes

Teacher action: Collect W4 as an independent assessment. Do not display its solution or substitute a hinted answer for an independent attempt.

Stage: 38 to 45 minutes

Teacher action: Discuss one anonymised explanation and set W5 as extension or the next opening task. Record which prerequisite needs reteaching.

Reduce help deliberately

Completion task: Use the photo-library case from the lesson. Classify the prediction, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

Teacher check: Prediction is learned because it was fitted using labelled pictures. Album placement is a written rule; the owner move is human control. Inspect the stored prediction and the placement or move record. A wrong album alone cannot distinguish prediction error from a later action.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Evidence and decision mechanisms. Proposed explanation: “All rules are outside AI”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Explain that AI also includes symbolic reasoning. In the specific bus case, the simple colour threshold is automation; the full system contains a fitted predictor.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Point to the learned stage. Does the changed display prove the weather forecast changed?

Reasoned teacher answer

The forecast stage is learned; it remains 70%. Only the boundary and resulting message changed.

Changed-case reasoning: The display no longer says TAKE ONE. 70 is below 80; changing a display rule does not refit the model.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Sort six devices by mechanism and defend two borderline cases.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR Talking proves that a machine learned.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

All rules are outside AI: Explain that AI also includes symbolic reasoning. In the specific bus case, the simple colour threshold is automation; the full system contains a fitted predictor.

A correct output proves a learned model: Ask whether a calculator, recording or person could produce the same correct output. Require evidence about fitting or reasoning.

Unknown is a bad answer: Accept unknown when the learner identifies a specific missing fact and a useful way to obtain it.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

The lamp uses a fixed rule; the photo sorter uses a learned model; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need mechanism details. Products can combine

mechanisms. A robot body or smart label proves none of them.

Independent question

A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a learned model? What evidence supports each classification?

Expected independent answer

The photo app uses a learned model because training examples shaped its behaviour. The toy is human-controlled; the clock follows fixed instructions. Speech and movement alone do not establish AI.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Talking proves that a machine learned.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

The prediction uses a learned model; the colour uses a written threshold; the override is human control. The red display alone reveals no training evidence. Ask how the delay estimate was produced. One product can combine mechanisms; simple automation does not define all of AI.

Explanations for the additional workbook examples

W1 Separate observations from explanations

A door opens when a visitor approaches. Its label says intelligent entrance. List two observations and write one mechanism question whose answer would help classify the opening decision.

Observations include that the door opened when the visitor approached and that its label says intelligent entrance. A useful question asks whether a written sensor threshold, a model fitted on examples or a person controls opening. The label is an observed marketing claim, not mechanism evidence.

W2 Complete the photo library explanation

Use the photo-library case from the lesson. Classify the prediction, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

Prediction is learned because it was fitted using labelled pictures. Album placement is a written rule; the owner move is human control. Inspect the stored prediction and the placement or move record. A wrong album alone cannot distinguish prediction error from a later action.

W3 Challenge an unsupported conclusion

Maya says a vacuum must learn because it changes direction after hitting a chair. Give two different mechanisms consistent with that observation. What evidence would distinguish them?

A written contact-sensor rule could turn the vacuum, or a learned navigation model could choose a turn. A person could also control it. Documentation of the route-selection process or a controlled trace could distinguish these mechanisms. Direction change alone is insufficient.

W4 Investigate a new hybrid system

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A

written rule displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent. Identify each mechanism and explain why a correct CHECK PLANTS message would not prove an accurate temperature prediction.

The forecast is learned-model prediction; the message is a written threshold; vent opening is the caretaker's decision. The message may correctly follow a forecast that is itself inaccurate. Compare the forecast with a predefined temperature observation at the relevant place and time before judging prediction accuracy.

W5 Design an evidence card

Invent one product containing two mechanisms. State the supplied facts, draw an information path and write one claim the facts support and one they do not. Do not use real classmates' personal information.

Accept a coherent case such as a trained speech recogniser supplying text to a written keyword rule. The support must connect to supplied facts, and the unsupported claim must exceed them, such as assuming perfect recognition. Check that the learner distinguishes the two decisions rather than merely naming two devices.

Assess independent transfer

Selected case: A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A written rule displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent. Identify each mechanism and explain why a correct CHECK PLANTS message would not prove an accurate temperature prediction.

Selected case answer

The forecast is learned-model prediction; the message is a written threshold; vent opening is the caretaker's decision. The message may correctly follow a forecast that is itself inaccurate. Compare the forecast with a predefined temperature observation at the relevant place and time before judging prediction accuracy.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or learned model; justify the smallest safe solution.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 2: Input, Process, Output. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Read the cases aloud and let students use arrows or spoken explanations. Supply the three mechanism labels if vocabulary is the barrier, but record that support. Extension students should design two

mechanisms producing the same output and propose evidence that would distinguish them.

For W4, require all three components classified with their supplied evidence and an explanation that a downstream rule can operate correctly on a wrong estimate. A label without supporting evidence is developing. If the learner says the display proves accuracy, return to the two bus rows and ask what each row actually checks. After feedback, assess a fresh case rather than awarding independent mastery for the corrected W4 answer.

Week 2 follows how information is represented and changed between components. Keep the three-row bus diagram: the delay number and the warning colour are different representations of related information.

Plan the delayed return

Suggested return points in this level: ai-1/2, ai-1/4, ai-1/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.